

No Primitive Higgs Scale: Carrier-Indexed Scale Standing and Hierarchy Orientation

Paper 3 of the Higgs Structural Endpoint Arc

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Abstract

This manuscript proves the third paper in the Higgs structural endpoint arc. Paper 1 established the Higgs bridge/fixation carrier

$$C_H := H_{\text{br/fix}}$$

as the standing-bearing carrier forced by the Higgs-domain order-to-deformation and mass-response target. Paper 2 proved the structural identity endpoint: Higgs identity is unique admissible occupation of C_H , not scalarity, vev-status, potential-minimum status, pole-mass status, RG-running status, vacuum-branch status, thermal status, or metric-comparison status. The present paper applies that carrier and identity package to Higgs-associated scale claims. It proves that there is no primitive Higgs scale independent of C_H .

The result does not deny measured Higgs-associated scales, renormalization-group scale dependence, electroweak input reconstruction, vacuum-instability diagnostics, cutoff sensitivity, or hierarchy comparisons. It denies their promotion into primitive Higgs-scale standing. The paper separates two questions that are often collapsed: the raw scale role of a displayed quantity and the standing status of the scale claim. A measured pole mass, a running coordinate, a cutoff display, an instability marker, or a metric ratio may be the raw role of a submitted scale expression; none of those roles is by itself a standing status. A scale claim has standing only when it is carrier-indexed: it must preserve the Higgs identity carrier C_H , the comparison domain, the admissibility boundary, and the relevant projection, transport, measurement, or route certificate. The standing scale normal form is

$$R_H^{\text{scale}} = (D_H, C_H, \mathcal{C}_{\text{comp}}, q, \Pi, \text{RawRole}_H^{\text{scale}}, \text{StandStatus}_H^{\text{scale}}, \text{Cert}),$$

where q is the scale representative or comparison object, $\mathcal{C}_{\text{comp}}$ is the declared comparison class, Π is the projection or transport map, $\text{RawRole}_H^{\text{scale}}$ records the descriptive role of q , and $\text{StandStatus}_H^{\text{scale}}$ records whether the submitted claim is carrier-indexed, standing-smuggling, target-updating, or a failed primitive assertion.

The main theorem is the No Primitive Higgs Scale Theorem:

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Scale}_H) = 0.$$

Equivalently, every admissible Higgs-scale claim factors through the carrier-indexed scale normal form; if the claim does not factor through C_H , it is representative-only, route-like, boundary-conventional, a metric projection, a target update, or a failed primitive claim. The hierarchy question is therefore not primarily a comparison between free-floating numbers such as v_{EW} and $\bar{M}_{\text{Pl}}$. It is a carrier-faithful comparison problem. Metricity and gravity enter only as downstream projection surfaces: a faithful metric projection can preserve a carrier-order relation, but it cannot generate that relation or convert a representative scale into a primitive Higgs ground. Under a faithful hierarchy projection, the Higgs carrier occupies the interior/subordinate side of the comparison. A reversed primitive scale reading either breaks carrier indexing, imports metric projection as authority, or changes target.

Keywords: AASC; Higgs scale; no primitive Higgs scale; carrier-indexed scale; hierarchy problem; Higgs bridge; Higgs structural identity; scale standing; metric projection; representative collapse; Standard Model.

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Mathematical-status lock and proof-class boundary

This paper is an AASC endpoint-classification theorem, not a numerical Standard Model calculation, a new hierarchy mechanism, a model-building mechanism, a UV completion, or a replacement for renormalized electroweak physics. Its target is the standing status of Higgs-associated scale claims after Papers 1 and 2 have fixed the Higgs bridge/fixation carrier C_H and the structural identity endpoint $\text{Id}_H = \text{Occ}_{D_H}(C_H)$.

The proof class is local and downstream. The kernel and consequence machinery are not rederived. Paper 1 imports the kernel, constructs the Higgs-domain carrier package, and proves carrier necessity. Paper 2 proves that Higgs identity is unique admissible occupation of C_H . The present paper asks which scale claims can carry standing once those results are in force. A same-target objection must therefore attack one of the following: the inherited carrier package, the inherited identity endpoint, the scale-candidate definitions, the raw-role and standing-status classifiers, the carrier-indexing normal form, the false-anchor eliminations, the conditional projection-orientation proof, or the endpoint survivor inference.

Read this first: proof spine

The proof has six layers.

Layer I: inherited carrier and identity. The paper works downstream of

$$C_H := H_{\text{br/fix}}, \quad \text{Id}_H = \text{Occ}_{D_H}(C_H).$$

No scale claim can determine what the Higgs is. It can only describe, measure, transport, compare, or project a scale associated with the already-fixed carrier.

Layer II: scale candidates and scale representatives. The familiar candidates

$$v_{\text{EW}}, m_H, m_H(\mu), \lambda(\mu), \Lambda_{\text{EW}}, \Lambda_{\text{cut}}, \Lambda_{\text{inst}}, M_{\text{Pl}}, \rho_{H/G}$$

are admitted as useful representatives or comparison objects. They are not admitted as primitive Higgs-scale grounds.

Layer III: scale normal form. An admissible Higgs-scale claim must factor through

$$R_H^{\text{scale}} = (D_H, C_H, \mathcal{C}_{\text{comp}}, q, \Pi, \text{RawRole}_H^{\text{scale}}, \text{StandStatus}_H^{\text{scale}}, \text{Cert}).$$

The normal form records carrier identity, comparison class, representative or projected quantity, transport/projection, raw role, standing status, and any certificate required for standing closure.

Layer IV: raw-role and standing-status exhaustion. Every submitted same-scope scale claim receives a raw scale role and a standing-status classification. Raw roles record whether the expression functions as measurement representative, running route, boundary convention, route marker, metric projection, or representative display. Standing status records whether the submitted claim is carrier-indexed, standing-smuggling, target-updating, or a failed primitive assertion. These layers are distinct: a measured or running expression may appear inside a carrier-indexed claim, but the measurement or running label is not itself the standing status. No additional same-scope scale-function locus remains.

Layer V: no primitive scale. If a scale candidate is standing-bearing, it is carrier-indexed or certificate-mediated, hence not primitive. If it is not carrier-indexed, it fails as a Higgs-scale ground. Thus

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Scale}_H) = 0.$$

Layer VI: hierarchy orientation. Hierarchy comparison is a carrier-faithful comparison problem before it is a metric ratio display. Metric projection can display a carrier order but cannot generate it. Under faithful projection, the Higgs carrier occupies the interior/subordinate side of the comparison.

1 Introduction: scale standing before scale mismatch

The ordinary hierarchy question is often phrased as a mismatch between a weak or Higgs scale and a high gravitational, cutoff, or ultraviolet scale. That phrasing is not wrong as physical language. It is incomplete as an endpoint claim. A scale mismatch is meaningful only after the compared quantities have standing in the same comparison mode. A fitted electroweak scale, a pole mass, a running parameter, a cutoff term, an instability crossing, and a Planck label do not have the same standing merely because each can be written as a number with dimensions.

The Higgs structural endpoint arc has already fixed the prior question. Paper 1 established the Higgs bridge/fixation carrier C_H as the standing-bearing object forced by the electroweak order-to-deformation and mass-response target. Paper 2 proved that Higgs identity is unique admissible occupation of that carrier. The present paper uses those results to classify scale language.

The central point is simple but demanding:

Higgs scale standing is carrier-indexed standing.

A Higgs-associated scale does not become primitive by being empirically measured, convenient in a Lagrangian, useful in a renormalization-group chart, diagnostic in an effective-potential analysis, or dramatic in a hierarchy display. It becomes standing-bearing only by preserving the inherited carrier identity and descending through a licensed measurement, transport, projection, or route certificate.

This paper therefore does not deny any of the following:

- the empirical usefulness of the electroweak scale;
- the physical importance of the Higgs excitation and pole mass;
- the role of running couplings and RG transport;
- the diagnostic force of cutoff sensitivity and heavy-threshold sensitivity;
- the meaningfulness of boundary-mode weak/gravity ratios;
- the value of hierarchy mechanisms in richer scopes.

It denies only the stronger primitive promotion:

$$q \Rightarrow \text{Stand}_{D_H}^{\text{prim}}(\text{Scale}_H),$$

where q is a scale representative not routed through the Higgs carrier and a standing certificate.

2 Dependency paragraph and inherited package

2.1 Local dependency posture

This paper works downstream of the Higgs carrier and identity package. It does not reprove the AASC kernel, the AMetric boundary, bivalence, standing conservation, fixed-domain exhaustion, or closure

by exhaustion. Those are imported through the Paper-1 local package and its support sources. The inherited package consists of:

$$D_H, \quad C_H, \quad H_{\text{role}}, \quad \mathcal{R}_H, \quad R_H, \quad \text{SF}_H, \quad E_H,$$

and the Paper-2 identity endpoint

$$\text{Id}_H = \text{Occ}_{D_H}(C_H).$$

Table 1: Inherited package used by Paper 3.

Imported item	Source role	Use in this paper
D_H	Same-scope Higgs-domain regime	Fixes the domain in which a scale claim is evaluated.
C_H	Higgs bridge/fixation carrier	Provides the required carrier for scale standing.
H_{role}	Gauge inheritance, deformation mediation, mass-response, identity continuity, standing transfer, redescription stability	Supplies the role bundle that scale representatives must preserve.
$\mathcal{R}_H$	Catalogue of scalar, vev, potential, mass, RG, vacuum, thermal, and metric descriptions	Supplies the representative classes tested as scale candidates.
R_H	Local Higgs normal form from Paper 1	Provides the broad factorization route for later Higgs claims.
$\text{Id}_H = \text{Occ}_{D_H}(C_H)$	Paper-2 identity endpoint	Blocks scale claims from determining Higgs identity.
Fixed-domain exhaustion	No fifth governance-bearing locus inside a fixed comparison class	Supports the raw-role/standing-status classifier and no-additional-class result.
No-generator/no-transfer/no-repair package	Standing cannot be generated by technical success, carried across domains, or repaired post hoc	Blocks primitive scale smuggling by measurement, RG transport, or metric display.

2.2 No re-foundation of the Higgs

A scale claim is not an identity claim. A scale claim can be measured, transported, compared, projected, bounded, or classified. It cannot re-open what counts as the Higgs unless it openly declares a target update. Thus every scale theorem below is conditioned on the inherited identity:

$$\text{Higgs} = \text{Occ}_{D_H}(C_H).$$

If a submitted scale claim changes the carrier, the comparison class, or the identity role, it is not a scale refinement of the same target. It is a target update.

3 What counts as a Higgs scale claim

Definition 3.1 (Higgs scale claim). A Higgs scale claim is a claim $\text{Sc}_H(q)$ that a quantity, relation, representative, running value, boundary parameter, crossing, or projection q fixes, anchors, compares, explains, or classifies a Higgs-associated scale inside the inherited domain D_H .

Definition 3.2 (Primitive Higgs scale). A primitive Higgs scale is a submitted scale claim q that purports to carry same-scope Higgs-scale standing directly as a scale ground, rather than as a measurement representative, transport coordinate, route marker, boundary convention, projection display, carrier-indexed certificate, or target update. It is denoted $\text{PrimScale}_H(q)$. This definition is neutral with respect to the later normal-form theorem: it identifies the claim-form under test, not its failure condition.

Definition 3.3 (Carrier-indexed scale). A carrier-indexed scale is a scale representative or comparison quantity whose standing is inherited through the Higgs carrier:

$$\text{CarrierIndexed}(q) \quad \text{only if} \quad q \text{ preserves } (D_H, C_H, H_{\text{role}})$$

and enters an admissible measurement, running, route, or projection normal form. A carrier-indexed scale is not primitive; its standing is mediated by C_H .

Definition 3.4 (Scale-standing modes). The paper distinguishes the following scale modes.

1. *Descriptive scale*: a coordinate or notation used to present a Higgs-associated quantity.
2. *Measured scale*: an empirical witness or reconstructed physical input.
3. *Running scale*: a renormalization or transport coordinate inside a declared scheme.
4. *Route marker*: a diagnostic crossing, threshold, branch, or transition marker used in a later endpoint route.
5. *Projection scale*: a display of carrier order or comparison status under a declared projection.
6. *Standing scale*: a carrier-indexed scale claim whose comparison class and certificate close.

Only the last mode carries standing closure. None is primitive independently of C_H .

Criterion 3.5 (Scale standing test). A submitted Higgs scale claim $\text{Sc}_H(q)$ has standing only if all of the following are fixed:

1. carrier preservation: q does not alter C_H or the identity endpoint;
2. comparison class: the domain in which the scale is compared is declared;
3. transport or projection: the map from representative to standing use is specified;
4. boundary status: empirical input, running coordinate, diagnostic marker, or projection mode is not confused with primitive standing;
5. certificate: any required uncertainty, scheme, quotient, route, or witness burden is declared.

4 Scale candidate catalogue

The following catalogue is admitted as useful physics language. Admission is not selection. The table records each scale candidate's lawful use and its primitive failure mode.

Table 2: Higgs-associated scale candidates and primitive failure modes.

Candidate	Lawful use	Standing status	Primitive failure mode
v_{EW} or vev-like coordinate	Electroweak normalization, potential presentation, symmetry-reduction coordinate	Measurement or representation indexed to C_H	Does not by itself supply carrier identity or scale standing; depends on input scheme or representation context.
m_H	Higgs excitation/pole response	Empirical response witness	Downstream witness of the carrier, not the carrier's primitive scale ground.
$m_H(\mu)$	Running mass coordinate in a scheme	RG transport coordinate	Changes with scale/scheme; standing comes from transport ledger, not chart value.
$\lambda(\mu)$	Running quartic, stability diagnostic	RG/route diagnostic	A crossing or value is not a Higgs-scale endpoint without quotient descent.
Λ_{EW}	Electroweak response scale shorthand	Boundary-normalized or response-class shorthand	Collapses into measured response unless internally certified.
Λ_{cut}	Regulator or EFT sensitivity display	Boundary convention or diagnostic	Cutoff choice is not a carrier-indexed scale anchor.
Λ_{match}	Threshold or EFT matching ledger	Transport datum	Matching split is not a primitive scale unless witnessed.
Λ_{inst}	Effective-potential instability/crossing marker	Vacuum-route diagnostic	Gauge/scheme/field-coordinate dependent unless mapped into a verdict route.
$M_{Pl}, \overline{M}_{Pl}$	Gravitational normalization display	Boundary display or metric projection ingredient	Planck label is not a Higgs carrier nor a denominator certificate by itself.
$\rho_{H/G}$	Weak/gravity comparison ratio	Boundary, mixed, or projection ratio	Meaningful only after numerator, denominator, and projection mode are typed.

Lemma 4.1 (Scale-candidate admission is not scale standing). *Every candidate in Table 2 may be admitted as descriptive, empirical, transport, diagnostic, or projection content. None has primitive Higgs-scale standing merely by being admitted.*

Proof. Admission says that a quantity can be written, measured, transported, or used in a calculation. Standing requires that the quantity preserve D_H , C_H , H_{role} , and the admissibility boundary under the role appropriate to the claim. Each listed candidate lacks at least one component when used primitively: vev-like quantities lack carrier priority; pole mass is a downstream witness; running quantities depend on transport; cutoffs and thresholds require boundary ledgers; instability crossings require route verdicts; metric labels require projection and denominator certificates. Therefore admission does not imply primitive standing. $\square$

5 Scale as representative

Definition 5.1 (Scale representative). A scale representative is a scale-associated expression q whose value or display is useful inside a presentation of D_H , but whose standing depends on carrier-indexed descent. The class of scale representatives is denoted $\mathcal{R}_H^{\text{scale}}$.

Lemma 5.2 (Scale-representative lemma). *Every Higgs-associated scale candidate in Table 2 is either a measurement representative, a renormalization or matching representative, a boundary convention, a route marker, or a projection quantity indexed to C_H .*

Proof. The table exhausts the ordinary scale roles used in the present target. A vev-like coordinate and the Higgs pole mass are measurement or response representatives. Running mass, running quartic, and matching scales are transport representatives. Cutoffs and Planck labels are boundary conventions unless a richer certificate is supplied. Instability crossings are route markers for vacuum analysis. Hierarchy ratios are projection quantities unless both carriers and the projection map are standing-bearing. In each case the representative is meaningful only relative to the inherited Higgs carrier; none fixes scale standing independently. $\square$

Proposition 5.3 (Representative preservation of physics). *Classifying a Higgs-associated scale as representative does not deny its physical use. It denies only its promotion into primitive same-scope scale standing.*

Proof. A representative may be indispensable for computation and measurement. The standing question asks whether it can ground a Higgs scale without preserving the carrier and comparison class. Since the paper preserves measurement, RG transport, matching, diagnostics, and projection modes while blocking only primitive promotion, no ordinary physical use is erased by representative classification. $\square$

6 Scale normal form

Definition 6.1 (Scale normal form). The carrier-indexed Higgs-scale normal form is

$$R_H^{\text{scale}} = (D_H, C_H, \mathcal{C}_{\text{comp}}, q, \Pi, \text{RawRole}_H^{\text{scale}}, \text{StandStatus}_H^{\text{scale}}, \text{Cert}),$$

where:

- D_H is the inherited Higgs-domain regime;
- C_H is the Higgs bridge/fixation carrier;
- $\mathcal{C}_{\text{comp}}$ is the declared comparison class;
- q is the scale representative or comparison object;
- Π is the measurement, transport, route, or projection map by which q is used;
- $\text{RawRole}_H^{\text{scale}}$ is the raw descriptive role of q , such as measurement representative, running route, boundary convention, route marker, metric projection, or representative display;
- $\text{StandStatus}_H^{\text{scale}}$ is the standing-status classification of the submitted claim;
- Cert is the certificate required for standing closure in the relevant mode.

Definition 6.2 (Classificatory and standing normalizers). The partial classificatory normalizer $\text{NF}_H^{\text{status}}(q)$ is defined when the submitted claim is typed inside D_H , preserves or explicitly updates the comparison target, and receives both a raw role and a standing-status classification. It records how the claim is classified, including failed, boundary, projection, and target-update outcomes.

The partial standing normalizer $\text{NF}_H^{\text{stand}}(q)$ is defined exactly when the submitted claim preserves D_H , preserves C_H , declares $\mathcal{C}_{\text{comp}}$, supplies Π , has standing status CarrierIndexed , and provides the certificate required for the asserted level of standing. When defined,

$$\text{NF}_H^{\text{stand}}(q) = R_H^{\text{scale}}(q).$$

The notation $\text{NF}_H^{\text{scale}}(q)$ is reserved for standing-normal use in theorem statements and means $\text{NF}_H^{\text{stand}}(q)$.

Definition 6.3 (Scale-form equivalence). For two standing-bearing scale claims q_1, q_2 , write

$$q_1 \sim_H^{\text{scale}} q_2$$

when they preserve the same carrier, comparison class, projection/transport route, raw role, standing-status classification, and certificate burden up to lawful redescription. Scale-form equivalence is stronger than numerical similarity and weaker than syntactic identity.

Theorem 6.4 (Scale normal form theorem). *Every standing-bearing same-scope Higgs scale claim factors through the standing scale normal form. More explicitly,*

$$\text{Stand}_{D_H}(\text{Sc}_H(q)) = 1 \Rightarrow \text{NF}_H^{\text{stand}}(q) \downarrow \wedge \text{StandStatus}_H^{\text{scale}}(q) = \text{CarrierIndexed} \wedge q \sim_H^{\text{scale}} \text{NF}_H^{\text{stand}}(q).$$

Proof. Standing-bearing scale use must preserve the inherited Higgs identity and cannot alter C_H . It must also declare what is being compared or transported; otherwise the scale is only a displayed representative. It must identify the map by which a representative becomes measurement, transport, route, or projection content. Finally, it must carry the certificate appropriate to its claim type. These fields are precisely the standing-normal fields of R_H^{scale} . If the claim is merely measured, running, boundary-conventional, diagnostic, or projected without carrier-indexed closure, $\text{NF}_H^{\text{status}}(q)$ may be defined as a classification record, but $\text{NF}_H^{\text{stand}}(q)$ is not defined. Therefore standing-bearing same-scope scale claims, and only those, factor through the standing normal form. $\square$

7 Scale classifier

Definition 7.1 (Raw scale-role classifier). The raw scale-role classifier records the descriptive work performed by the submitted expression before standing status is assigned:

$$\text{RawRole}_H^{\text{scale}}(q) \in \{\text{MeasurementRep}, \text{RunningRoute}, \text{BoundaryConvention}, \text{RouteMarker}, \text{MetricProjection}, \text{RepresentativeMode}\}.$$

Its classes are:

1. **MeasurementRep**: empirical or reconstructed response representative, including v_{EW} and m_H ;
2. **RunningRoute**: RG, scheme, matching, or transport coordinate;
3. **BoundaryConvention**: cutoff, Planck label, matching choice, input convention, or threshold split treated as a boundary datum;
4. **RouteMarker**: vacuum-instability crossing, lower-branch diagnostic, thermal marker, or other downstream route signal;
5. **MetricProjection**: metric/gravity display or ratio projection;
6. **RepresentativeMode**: scalar-potential, vev-like, field-coordinate, or notation-level presentation not covered by the preceding roles.

Definition 7.2 (Standing-status classifier). The standing-status classifier records the governance status of the submitted scale claim:

$$\text{StandStatus}_H^{\text{scale}}(q) \in \{\text{CarrierIndexed}, \text{StandingSmuggling}, \text{TargetUpdate}, \text{FailedPrimitive}\}.$$

Its classes are:

1. **CarrierIndexed**: the claim preserves D_H and C_H , declares $\mathcal{C}_{\text{comp}}$, supplies Π , and carries the stated certificate burden;
2. **StandingSmuggling**: a representative, route, measurement, projection, or technical success is used as scale authority before carrier-indexed closure;
3. **TargetUpdate**: the claim changes carrier, comparison class, source inventory, or admissibility boundary;
4. **FailedPrimitive**: the claim is offered as a primitive Higgs-scale ground and supplies no carrier-indexed standing route.

Remark 7.3 (Descriptive role is not standing status). A single submitted claim has both a raw role and a standing status. A pole-mass claim may have raw role **MeasurementRep**; it is standing-bearing only if its standing status is **CarrierIndexed**. An RG scale claim may have raw role **RunningRoute**; it is standing-bearing only through a carrier-indexed transport certificate. Thus measurement, running, boundary, route, and projection modes do not compete with **CarrierIndexed**; they are descriptive modes that may appear inside, or fail to reach, carrier-indexed standing.

Table 3: Raw scale functions and fixed-domain loci.

Scale function	Fixed-domain locus	Raw role	Standing consequence
Measure	Empirical witness / standing partition	MeasurementRep	May enter carrier-indexed standing as witness; never primitive by measurement alone.
Represent	Lawful quotient / skin	RepresentativeMode	Presentation content; requires descent to standing normal form.
Transport	Admissible continuation	RunningRoute	Route or RG content; standing requires transport certificate.
Diagnose	Route marker / standing partition	RouteMarker	Diagnostic only unless routed through an endpoint or scale certificate.
Compare	Projection / comparison class	MetricProjection	Projection display; cannot generate carrier order.
Conventionalize	Boundary typing	BoundaryConvention	Boundary datum; no primitive standing.
Select	Standing partition / no-generator discipline	no lawful raw role as primitive selector	Standing smuggling unless independently certified through carrier normal form.
Update	Boundary typing / domain identity	target update	Richer-scope mechanism, not same-scope primitive scale.

Table 4: Standing-status classes and fixed-domain loci.

Standing status	Fixed-domain locus	Normalizer	Reason no further standing class remains
CarrierIndexed	Lawful quotient / admissible continuation	$\text{NF}_H^{\text{stand}}(q) \downarrow$	The claim preserves carrier identity and supplies the declared scale route and certificate.
StandingSmuggling	Standing partition / no-generator discipline	$\text{NF}_H^{\text{status}}(q) \downarrow$, $\text{NF}_H^{\text{stand}}(q) \uparrow$	The claim imports authority before carrier-indexing or closure.

Standing status	Fixed-domain locus	Normalizer	Reason no further standing class remains
TargetUpdate	Boundary typing / domain identity	$\text{NF}_H^{\text{status}}(q) \downarrow$, different target	The claim changes the target rather than refining scale standing.
FailedPrimitive	Standing partition	$\text{NF}_H^{\text{status}}(q) \downarrow$, $\text{NF}_H^{\text{stand}}(q) \uparrow$	Primitive scale assertion lacks standing-normal fields.

Theorem 7.4 (Raw-role and standing-status totality). *Every submitted same-scope Higgs scale claim receives a raw role and a standing-status classification. No additional same-scope scale-function locus remains.*

Proof. Let q be a submitted scale claim. As a scale expression, it must measure, represent, transport, diagnose, compare, conventionalize, select, or update. These functions are the fixed-domain loci displayed in Table 3: empirical witness, lawful quotient/skin, admissible continuation, route marker, projection/comparison class, boundary typing, standing partition, and domain update. The first six functions assign raw roles. Selection is not a lawful raw role for primitive authority; it is standing smuggling unless it is independently certified through the carrier normal form. Update changes the target. Thus every submitted claim receives a raw role or is identified as update/selector behavior.

For standing status, if q preserves D_H and C_H , declares $\mathcal{C}_{\text{comp}}$, supplies Π , and provides the required certificate, then $\text{StandStatus}_H^{\text{scale}}(q) = \text{CarrierIndexed}$. If it uses a representative, route, measurement, projection, or technical success as authority before such closure, it is **StandingSmuggling**. If it changes carrier, comparison class, or source inventory, it is **TargetUpdate**. If it asserts primitive Higgs-scale standing without carrier-indexed fields, it is **FailedPrimitive**. A further same-scope standing class would have to change scale-standing authority while altering none of boundary typing, standing partition, lawful quotient, admissible continuation, projection/comparison mode, or domain update. Fixed-domain exhaustion excludes that possibility. $\square$

8 Primitive Higgs scale collapse

Lemma 8.1 (Scale-function exhaustion). *A submitted Higgs scale claim must perform one of the scale functions in Table 3: measure, represent, transport, diagnose, compare, conventionalize, select, or update. A further asserted same-scope function is bookkeeping unless it alters one of the fixed-domain loci already listed.*

Proof. A scale claim can enter the target only by doing some scale work. If it reports a measured or reconstructed value, it measures. If it displays a field coordinate, potential coordinate, vev-like coordinate, or formal scalar expression, it represents. If it moves through RG, scheme, matching, or transport data, it transports. If it marks a vacuum crossing, threshold, route status, or transition signal, it diagnoses. If it forms a ratio or hierarchy display, it compares. If it fixes a cutoff, label, input set, matching split, or unit convention, it conventionalizes. If it chooses one scale as authoritative, it selects. If it changes the carrier, source inventory, or comparison class, it updates the target. These exhaust scale-standing work because any scale use that changes standing must alter empirical witness status, lawful quotient content, admissible continuation, route marking, comparison/projection,

boundary typing, standing selection, or target identity. A remaining expression changes none of those loci and is bookkeeping. $\square$

Theorem 8.2 (No Primitive Higgs Scale Theorem). *Inside the inherited Higgs-domain regime D_H , no Higgs-associated scale has primitive same-scope standing independent of C_H :*

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Scale}_H) = 0.$$

Equivalently,

$$\text{PrimScale}_H(q) \Rightarrow \text{Stand}_{D_H}^{\text{prim}}(q) = 0,$$

unless the submitted claim is rerouted through $\text{NF}_H^{\text{stand}}(q) \downarrow$, in which case it is carrier-indexed rather than primitive.

Proof. Take any submitted primitive scale claim q . By the scale-function exhaustion lemma, q measures, represents, transports, diagnoses, compares, conventionalizes, selects, or updates. If it measures, it supplies an empirical or reconstructed response representative; measurement does not by itself ground the Higgs carrier. If it represents, it supplies field-coordinate, potential, vev-like, or scalar-chart content; representative content requires quotient and carrier descent. If it transports, it is an RG, scheme, or matching route; transport standing requires a transport certificate. If it diagnoses, it marks a vacuum crossing, instability scale, branch, threshold, or route status; diagnostic content requires an endpoint or route certificate. If it compares, it is a ratio or projection display; comparison requires carrier and projection typing. If it conventionalizes, it is a boundary input, cutoff, Planck label, matching split, or unit choice; boundary convention is not primitive scale standing. If it selects, it imports standing authority before carrier-indexed closure and is standing smuggling. If it updates, it changes the target. Thus no same-scope function supplies primitive Higgs-scale standing. When a scale claim has standing, it has $\text{NF}_H^{\text{stand}}(q) \downarrow$ and $\text{StandStatus}_H^{\text{scale}}(q) = \text{CarrierIndexed}$, so it is not primitive. Therefore $\text{Surv}_{D_H}^{\text{prim}}(\text{Scale}_H) = 0$. $\square$

Corollary 8.3 (No vev primitive).

$$v_{\text{EW}} \not\Rightarrow \text{Stand}_{D_H}^{\text{prim}}(\text{Scale}_H).$$

A vev-like value can be a measured or reconstructed representative of electroweak response, but it is not the primitive Higgs scale.

Corollary 8.4 (No pole-mass primitive).

$$m_H \not\Rightarrow \text{Stand}_{D_H}^{\text{prim}}(\text{Scale}_H).$$

The Higgs pole mass is a response witness, not an identity-grounding scale anchor.

Corollary 8.5 (No RG-crossing primitive).

$$\lambda(\mu) = 0 \text{ or } \Lambda_{\text{inst}} \not\Rightarrow \text{Stand}_{D_H}^{\text{prim}}(\text{Scale}_H).$$

Running crossings and instability markers require transport and route verdicts before standing-bearing use.

9 False scale anchor exhaustion

Table 5: False scale anchors and AASC reduction.

False anchor	Attempted primitive promotion	AASC reduction
Observed vev	The weak scale is primitive because it is measured	Boundary-normalized response representative; standing depends on C_H .
Potential minimum	The potential minimum fixes the Higgs scale	Field-coordinate/potential representative; requires quotient and witness descent.
Bare mass term	Lagrangian scalar coordinate fixes scale origin	Scheme and boundary dependent; not a scale anchor by itself.
Cutoff correction	Large Λ_{cut}^2 -sensitivity fixes hierarchy target	Sensitivity diagnostic; not a selector without quotient-stable criterion.
Heavy threshold split	Threshold contribution explains or selects scale	Matching ledger; split needs witness or richer-scope mechanism.
Running zero/crossing	RG crossing is the scale endpoint	Transport/diagnostic coordinate; not standing without route certificate.
Instability scale	Field value of vacuum instability is physical scale	Gauge/scheme/field-coordinate diagnostic unless mapped to endpoint verdict.
Planck label	M_{Pl} or $\overline{M}_{\text{Pl}}$ is the denominator anchor	Gravitational boundary display; requires carrier/projection certificate.
Raw ratio	$v_{\text{EW}}/\overline{M}_{\text{Pl}}$ solves or defines hierarchy	Boundary-ratio mode unless both scale maps are derived and carrier-faithful.
Naturalness score	Sensitivity ranking selects scale	Heuristic or diagnostic unless quotient-stable and carrier-indexed.

Theorem 9.1 (False Scale Anchor Exhaustion). *Every attempted primitive Higgs scale anchor in Table 5 receives a raw scale role and a standing-status classification. Its raw role may be measurement, representation, transport, route marker, boundary convention, or projection; its same-scope standing status is carrier-indexed only after the standing normal form closes. No false anchor supplies same-scope primitive scale standing.*

Proof. Each row reduces to a raw scale role and then to a standing-status outcome. Vev, pole mass, and electroweak scale are measurement representatives. Potential minima are representative displays. Running values, RG crossings, and matching ledgers are transport roles or route markers. Instability field values are route markers for vacuum analysis. Cutoffs, Planck labels, and threshold splits are boundary conventions. Raw weak/gravity ratios are metric or boundary projections. Naturalness scores are diagnostic unless they carry independent quotient-stable authority, in which case that authority must be declared as a certificate or target update. None of these raw roles is primitive scale standing; only carrier-indexed standing through $\text{NF}_H^{\text{stand}}$ can survive. $\square$

10 Carrier ordering before metric ratio

Definition 10.1 (Carrier comparison). A carrier comparison is a relation between standing-bearing carriers, not between free-floating numerical representatives. Write

$$C_H \triangleleft C_X$$

when the Higgs carrier is evaluated as an interior or subordinate carrier relative to comparison carrier C_X in the declared comparison class.

Lemma 10.2 (Carrier-order priority). *An admissible hierarchy comparison must compare standing-bearing carriers before it reports numerical or metric display quantities.*

Proof. A numerical ratio is a function of a numerator scale map, a denominator scale map, and a comparison mode. If the carriers are not fixed first, the numerator and denominator can be changed by representatives, input choices, boundary labels, or target updates. A carrier comparison fixes what is being compared; numerical display can then be interpreted as measurement, boundary ratio, projection, or derived ratio. Therefore carrier order has priority over ratio display. $\square$

Proposition 10.3 (Hierarchy target is carrier-faithful). *A hierarchy claim about the Higgs is same-scope only if its Higgs-side term preserves C_H and its comparison-side term declares a carrier, boundary input, or projection surface.*

Proof. If the Higgs-side term fails to preserve C_H , the claim is no longer about the same Higgs identity. If the comparison-side term is undeclared, the ratio may compare a carrier with a label, a measurement with a convention, or a route marker with a boundary input. That ambiguity prevents standing. Carrier-faithful comparison fixes both sides before any scale display is reported. $\square$

11 Metric projection as downstream representative

Definition 11.1 (Metric projection). A metric projection is a map

$$\Pi_{\text{met}} : (C_H, C_X) \longrightarrow q_{\text{metric}}$$

that displays a carrier comparison by means of a metric, gravitational, Planck-normalized, closure-scale, or numerical ratio representative. The projection is faithful when it preserves the declared carrier order and does not import empirical or boundary labels as primitive authority.

Lemma 11.2 (Metric projection non-generation). *A metric projection can preserve a carrier-order relation, but it cannot generate the carrier order and cannot convert a representative scale into primitive Higgs standing.*

Proof. Projection is a display or transport of already fixed comparison content. If the carrier relation is absent, a metric projection must obtain its ordering from an independent selector, empirical boundary, or convention. That would be standing smuggling or target update. If the carrier relation is present, projection may faithfully display it, but the standing comes from the carrier comparison and projection certificate. Thus projection preserves but does not generate. $\square$

Corollary 11.3 (Metricity as downstream skin). *Metricity is not foundational for the Higgs scale theorem. It is a downstream projection surface whose scale role is admitted only after carrier indexing.*

12 Conditional faithful-projection orientation theorem

Definition 12.1 (Faithful hierarchy projection). A faithful hierarchy projection for the Higgs target is a metric or closure-scale projection Π_{met} satisfying:

1. carrier descent: it is defined on carriers, not bare representatives;
2. order preservation: it preserves the declared carrier order;
3. calibration firewall: it does not import v_{EW} , $\overline{M}_{\text{Pl}}$, or $\rho_{H/G}$ as primitive authority;
4. target stability: it does not change D_H , C_H , or the comparison carrier.

Theorem 12.2 (Conditional Faithful-Projection Orientation Theorem). *Assume the declared carrier-order premise $C_H \triangleleft C_G^{\text{env}}$ supplied by the scale-envelope support route for the Higgs/metric comparison. Under a faithful hierarchy projection, an admissible Higgs hierarchy comparison displays C_H on the interior or subordinate side of the comparison. In notation,*

$$\text{Faithful}(\Pi_{\text{met}}) \wedge C_H \triangleleft C_G^{\text{env}} \Rightarrow 0 < \Pi_{\text{met}}(C_H) < \Pi_{\text{met}}(C_G^{\text{env}})$$

when the projection reports a positive ordered scale display. A reversed primitive reading either breaks carrier indexing, treats metric projection as primitive authority, or changes target.

Proof. The carrier relation $C_H \triangleleft C_G^{\text{env}}$ is a relation between standing-bearing carriers. A faithful projection preserves that relation and does not import a boundary display as authority. Therefore, when the projection is positive and order-reporting, it places the Higgs carrier on the interior/subordinate side of the display. If a submitted reading reverses the comparison while claiming the same target, then it either changes the carrier order, uses a metric display to generate rather than preserve order, or changes the comparison carrier. These are respectively carrier-indexing failure, standing smuggling, or target update. Hence no reversed same-scope primitive hierarchy reading survives. $\square$

Nonclaim 12.3 (No numerical hierarchy solution). The theorem does not prove a numerical value for $\rho_{H/G}$, does not derive v_{EW} , does not derive $\overline{M}_{\text{Pl}}$, does not select a UV completion, and does not independently derive the carrier-order premise $C_H \triangleleft C_G^{\text{env}}$. It closes only the scale-standing status and the conditional projection report for the declared carrier-faithful comparison.

13 Relation to standard hierarchy language

Standard hierarchy language remains meaningful in its proper mode. A boundary ratio such as $v_{\text{EW}}/\overline{M}_{\text{Pl}}$ compares measured or boundary-normalized quantities. A sensitivity argument using Λ_{cut} or heavy thresholds diagnoses instability or missing protection in a presentation. An RG argument transports scale dependence through a scheme and matching ledger. A UV mechanism may supply a new carrier or projection certificate in a richer scope.

The paper refuses only one move: treating those descriptions as primitive Higgs-scale standing without C_H and without a declared certificate. The conventional hierarchy problem is therefore preserved as a physical target but retyped as a carrier-faithful scale-standing problem.

Table 6: Common hierarchy formulations under the scale-standing classifier.

Formulation	Preserved content	Status in this paper
Weak/Planck smallness	Meaningful boundary comparison	Boundary or metric projection unless both carriers and scale maps are certified.
Quadratic sensitivity	EFT and regulator diagnostic	Sensitivity diagnosis; not primitive scale selector.
Heavy threshold correction	Genuine model/EFT pressure	Matching or richer-scope route; not same-scope primitive anchor.
Naturalness criterion	Heuristic and model-building pressure	Diagnostic unless quotient-stable and carrier-indexed.
Dimensional transmutation	Candidate mechanism for generated scale	Serious only if projected to C_H by a standing-bearing map.
SUSY or compositeness	Candidate richer-scope protection or replacement mechanism	Target update requiring carrier-preservation or projection certificate.
Metric-envelope order display	Structural projection mode	Downstream projection; preserves but does not found the Higgs scale theorem.

14 Endpoint register

Theorem 14.1 (Scale endpoint register). *The Paper-3 exhaustion register is:*

$$\begin{aligned}
\text{Surv}_{D_H}^{\text{prim}}(v_{\text{EW}}) &= 0, \\
\text{Surv}_{D_H}^{\text{prim}}(m_H) &= 0, \\
\text{Surv}_{D_H}^{\text{prim}}(\lambda(\mu)) &= 0, \\
\text{Surv}_{D_H}^{\text{prim}}(\Lambda_{\text{cut}}) &= 0, \\
\text{Surv}_{D_H}^{\text{prim}}(\Lambda_{\text{inst}}) &= 0, \\
\text{Surv}_{D_H}^{\text{prim}}(M_{\text{Pl}}) &= 0, \\
\text{Surv}_{D_H}^{\text{prim}}(\rho_{H/G}) &= 0, \\
\text{Surv}_{D_H}^{\text{prim}}(\text{Scale}_H) &= 0.
\end{aligned}$$

The surviving same-scope status is carrier-indexed scale standing through R_H^{scale} , not primitive scale standing.

Proof. The individual equations are corollaries of the false-scale-anchor and no-primitive-scale theorems. Each candidate has lawful use only as measurement representative, running route, boundary convention, route diagnostic, projection, or target update. Carrier-indexed scale standing survives because it factors through R_H^{scale} . Primitive scale standing has no surviving same-scope representative. $\square$

15 Endpoint theorem set

This section consolidates the proof outputs so the manuscript can be audited without reconstructing the whole chain from prose.

Theorem 15.1 (Scale-use governance). *Every claim-bearing Higgs-scale use in D_H presupposes the inherited carrier and identity package:*

$$\text{Sc}_H(q) \wedge \text{Stand}_{D_H}(q) = 1 \Rightarrow \text{Id}_H = \text{Occ}_{D_H}(C_H).$$

Proof. By Paper 2, Higgs identity in D_H is the unique admissible occupation of C_H . A scale claim about the Higgs that does not preserve this identity either changes target or fails as a same-scope Higgs claim. Hence standing-bearing scale use presupposes the inherited carrier and identity package. $\square$

Theorem 15.2 (Typed scale classification). *Every Higgs scale claim receives a raw-role and standing-status classification. Raw roles include measurement representative, running or transport route, route marker, boundary convention, metric projection, and representative display. Standing-status outcomes are carrier-indexed, standing-smuggling, target-update, or failed-primitive. Among same-scope standing-bearing scale claims, only carrier-indexed scale content survives.*

Proof. This is the raw-role and standing-status totality theorem applied after carrier identity is fixed. The classification is quotient-based because lawful redescription preserves the scale role and standing status while not necessarily preserving a literal formula or number. A scale claim outside the classifier changes none of the fixed-domain loci, in which case it is bookkeeping, or changes target, in which case it is a target update. Standing-bearing use requires $\text{NF}_H^{\text{stand}}(q) \downarrow$ and $\text{StandStatus}_H^{\text{scale}}(q) = \text{CarrierIndexed}$. $\square$

Theorem 15.3 (Primitive scale deletion). *The primitive Higgs scale claim-form has no same-scope survivor:*

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Scale}_H) = 0.$$

Proof. This restates the No Primitive Higgs Scale Theorem as an endpoint theorem. All same-scope scale candidates either factor through R_H^{scale} , in which case they are not primitive, or fail as primitive. Thus the primitive claim-form is deleted while carrier-indexed scale use survives. $\square$

Theorem 15.4 (Hierarchy orientation endpoint). *A same-scope hierarchy comparison that treats the Higgs term as primitive scale rather than carrier-indexed content is deleted. A faithful projection may display C_H as interior/subordinate relative to a declared comparison carrier, but it cannot generate Higgs-scale standing.*

Proof. The first sentence follows from primitive scale deletion. The second follows from metric projection non-generation and the Conditional Faithful-Projection Orientation Theorem. Projection reports the carrier comparison; it does not authorize a representative scale by itself. $\square$

16 Handoff to vacuum routes

The scale theorem prepares the vacuum-route paper. Vacuum-instability scales and effective-potential crossings are not primitive Higgs scales. They are diagnostic or route-status markers that require quotient, gauge, RG, matching, and lifetime or thermal verdict maps before endpoint use. Thus

Paper 4 can treat lower branches, metastability, lifetime routes, thermal transitions, and electroweak-baryogenesis route claims without allowing a displayed instability scale to become a primitive Higgs attribute.

17 Conclusion

Once Higgs identity is fixed by C_H , scale language is downstream. The vev, pole mass, running parameters, cutoff corrections, matching thresholds, instability crossings, Planck labels, and weak/gravity ratios all remain available as useful physics language. None is a primitive Higgs scale. The standing-bearing scale object is not a free-floating number; it is a carrier-indexed claim whose comparison class, projection or transport route, and certificate must be declared.

The endpoint is therefore:

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Scale}_H) = 0,$$

with same-scope scale standing surviving only through

$$R_H^{\text{scale}} = (D_H, C_H, \mathcal{C}_{\text{comp}}, q, \Pi, \text{RawRole}_H^{\text{scale}}, \text{StandStatus}_H^{\text{scale}}, \text{Cert}).$$

The hierarchy problem is sharpened rather than erased. It becomes a question of carrier-faithful comparison and faithful projection, not primitive scale mismatch. Metricity is a projection surface; it may display a carrier relation, but it does not ground the Higgs scale.

A Scale candidate register

Table 7: Scale candidate register.

Candidate	Raw role	Failed primitive criterion	Allowed role
v_{EW}	MeasurementRep	Carrier priority; empirical ancestry	Electroweak response representative
m_H	MeasurementRep	Downstream witness, not carrier anchor	Higgs excitation response
$m_H(\mu)$	RunningRoute	Scheme/scale dependence	RG transport coordinate
$\lambda(\mu)$	RunningRoute	Running chart does not close endpoint	Stability diagnostic
Λ_{EW}	MeasurementRep or BoundaryConvention	Boundary reconstruction if imported	Response shorthand
Λ_{cut}	BoundaryConvention	Regulator choice not standing anchor	Sensitivity diagnostic
Λ_{match}	BoundaryConvention or RunningRoute	Matching split not intrinsic	EFT transport ledger

Candidate	Raw role	Failed primitive criterion	Allowed role
Λ_{inst}	RunningRoute	Gauge/scheme/route incompleteness	Vacuum diagnostic
M_{Pl}	BoundaryConvention or MetricProjection	Denominator label not carrier certificate	Gravitational display
$\rho_{H/G}$	MetricProjection	Raw ratio lacks derivational ancestry	Boundary/projection ratio

B Classifier normal forms

Table 8: Raw scale-role normal forms.

Raw role	Normal-form condition	Standing implication	Allowed use
Measurement representative	Empirical or reconstructed response representative; carrier preserved	Not primitive; may be carrier-indexed with certificate	Physical witness
Running route	RG/scheme/matching route declared; carrier preserved	Transport does not close standing by itself	RG or matching ledger
Route marker	Crossing, branch, threshold, or thermal/vacuum marker	Diagnostic only without route verdict	Endpoint-route input
Boundary convention	Input, regulator, cutoff, Planck label, threshold split, or convention	Boundary status; no primitive standing	Convention or source datum
Metric projection	Metric/gravity display of carrier comparison	Projection status; does not generate order	Ratio or closure-scale display
Representative mode	Scalar, vev-like, potential, field-coordinate, or notation-level display	Presentation; requires quotient descent	Descriptive calculation mode

Table 9: Standing-status normal forms.

Standing status	Normal-form condition	Endpoint consequence
CarrierIndexed	$\text{NF}_H^{\text{stand}}(q) \downarrow$ with carrier, comparison, projection, raw role, standing status, and certificate fields declared	Eligible for standing-bearing scale use at the certified level.

Standing status	Normal-form condition	Endpoint consequence
StandingSmuggling	Representative, measurement, route, projection, or technical success used before closure	Fail-closed; no standing-normal form.
TargetUpdate	Carrier, comparison class, source inventory, or admissibility boundary changes	Richer-scope theorem required.
FailedPrimitive	Primitive scale asserted without standing-normal fields	Deleted as same-scope primitive.

C Comparison with common hierarchy formulations

The hierarchy literature often uses a mixed vocabulary of measured inputs, formal scalar coordinates, running quantities, regulator displays, and high-scale denominators. Paper 3 does not reject that language. It assigns each item a standing role. The weakest admissible report is boundary display; the strongest would be a quotient-singleton scale value derived without hidden empirical ancestry. The present paper closes neither numerical singleton nor finite suppression magnitude. It closes primitive-scale promotion and carrier-orientation status.

Definition C.1 (Boundary-ratio mode). A ratio is in boundary-ratio mode when both numerator and denominator are measured or boundary-normalized inputs. The display is physically meaningful but not a same-domain derivation.

Definition C.2 (Projection-ratio mode). A ratio is in projection-ratio mode when the numerator and denominator are standing-bearing carriers and a projection displays their comparison. The display inherits standing from the carrier relation and projection certificate.

Definition C.3 (Derived-ratio mode). A ratio is in derived-ratio mode when both scale maps and the comparison map are certified without importing the target values as empirical anchors. This paper does not claim derived-ratio closure.

D Formal statement of the optional hierarchy ratio

If an additional support theorem supplies a faithful closure-scale projection, one may write

$$\rho_{H/G}^{\text{proj}} = \frac{\Pi(C_H)}{\Pi(C_G^{\text{env}})}.$$

The present theorem permits the order-type report under faithful projection:

$$0 < \rho_{H/G}^{\text{proj}} < 1$$

only as a projection of a declared carrier-order comparison. It does not identify $\rho_{H/G}^{\text{proj}}$ with $v_{\text{EW}}/\overline{M}_{\text{Pl}}$ and does not determine a numerical magnitude.

E Denial-obligation register

A denial of the Paper-3 endpoint has same-target standing only if it does at least one of the following:

1. breaks the inherited Paper-1 carrier package;
2. breaks the Paper-2 identity endpoint $\text{Id}_H = \text{Occ}_{D_H}(C_H)$;
3. supplies a same-scope primitive scale that preserves C_H while not factoring through R_H^{scale} ;
4. identifies an additional scale-function or standing-status class not reducible to the fixed-domain loci of Table 4;
5. proves that a representative scale can carry standing without comparison class, projection/transport, or certificate fields;
6. shows that a metric projection can generate rather than display carrier order without standing smuggling;
7. supplies a faithful reversed hierarchy orientation that preserves the same carrier relation and target;
8. declares a richer-scope scale mechanism and accepts target update.

Absent one of these results, the objection changes target, treats a representative as a primitive, or refuses the AASC vocabulary while retaining the same governance work.

F Anti-circularity and forbidden-retyping audit

Table 10: Anti-circularity and forbidden-retyping audit.

Potential circularity or forbidden retyping	Audit response
D_H smuggles in scale deletion	The target fixes the inherited carrier and identity package; primitive scale deletion follows only after the scale catalogue, classifier, normal form, and false-anchor exhaustion.
C_H is defined as the scale anchor	False. C_H is the inherited bridge/fixation carrier. Scale claims must factor through it, but C_H is not itself a numerical scale value.
Carrier-indexing is a synonym for success	False. Carrier-indexing is a normal-form condition; standing closure still requires the comparison class, projection/transport route, and certificate appropriate to the claim.
Metric projection is assumed to be subordinate	False. Subordination is a carrier-order premise imported only for the declared hierarchy comparison; the paper proves what faithful projection may report, not a global physical-energy ordering.
Vev and pole mass are denied as physical	False. They are preserved as measurement or response representatives and denied only primitive scale authority.

Potential circularity or forbidden retyping	Audit response
Running scales are rejected	False. RG and matching quantities are admitted as transport routes and denied only primitive standing independent of route/certificate context.
The hierarchy problem is declared solved	False. The paper closes primitive-scale promotion and hierarchy orientation under faithful projection; it does not derive a numerical weak/Planck ratio.
BSM mechanisms are ruled out	False. New carriers, thresholds, symmetries, or UV dynamics are target updates requiring their own carrier and projection certificates.

G Standing-objection register

Table 11: Standing-objection register for Paper 3.

Objection	Standing	Response
The observed weak scale is physical.	Granted	It is a measurement representative; physical use does not make it primitive scale standing.
The Higgs pole mass is a real observable.	Granted	It is a response witness, not the primitive Higgs scale.
RG running is essential to Higgs stability.	Granted	Running is transport; transport does not by itself select a primitive scale.
Cutoff sensitivity motivates hierarchy mechanisms.	Granted	Sensitivity diagnoses missing structure; it is not same-scope scale selection without a certificate.
Planck comparison is physically meaningful.	Granted in boundary mode	The ratio is meaningful, but denominator standing requires carrier/projection typing for derivational mode.
Naturalness remains important.	Granted as heuristic or diagnostic	It becomes a scale selector only if quotient-stable, carrier-indexed, and calibration-prior.
Metric projection can display order.	Granted	Projection preserves declared carrier order; it does not generate the order.
A UV completion may generate the Higgs scale.	Compatible	That is a richer-scope mechanism requiring carrier-preservation and projection back to D_H .
A primitive scale must exist because numbers are measured.	No same-target force	Measurement supplies witnesses; primitive standing requires normal-form closure.

H Faithful same-regime counterexample conditions

A faithful same-target counterexample to Paper 3 must preserve all of the following:

1. the inherited Higgs-domain target D_H ;
2. the inherited carrier C_H ;
3. the identity endpoint $\text{Id}_H = \text{Occ}_{D_H}(C_H)$;
4. the same comparison class for the scale claim;
5. the same admissibility boundary and no-repair/no-transfer discipline;
6. the distinction between measurement, transport, boundary convention, route marker, and projection;
7. the raw-role and standing-status classifiers or governance-equivalent replacements;
8. the endpoint-selection standard distinguishing primitive from carrier-indexed standing.

While preserving those conditions, it must produce at least one of the following:

1. a same-scope primitive Higgs scale that does not factor through R_H^{scale} ;
2. an additional scale-function or standing-status class not reducible to the fixed-domain loci;
3. a metric projection that creates carrier order without standing smuggling;
4. a reversed hierarchy orientation that preserves the declared carrier relation.

A construction that changes carrier, comparison class, empirical ancestry, scale map, or admissibility boundary is a target update rather than a same-target counterexample.

I Right-mode objection targets

A right-mode objection must attack at least one load-bearing link:

1. the inherited Paper-1 carrier package;
2. the inherited Paper-2 identity endpoint;
3. the definition of Higgs scale claim;
4. the distinction between primitive, representative, route, boundary convention, and projection;
5. the scale normal form R_H^{scale} ;
6. the raw-role and standing-status classifiers and no-additional-scale-function exhaustion;
7. the false-scale-anchor reductions;
8. the no primitive Higgs scale theorem;
9. the carrier-order priority lemma;
10. the metric projection non-generation lemma;
11. the conditional faithful-projection orientation theorem;
12. the final endpoint register.

Objections outside these targets either change the subject, object to the AASC framework in general, or dispute a downstream physical preference without breaking the scale-standing theorem.

J Claim and nonclaim register

Table 12: Claim and nonclaim register.

ID	Statement	Status
S3-T1	Higgs scale claims are evaluated downstream of C_H and Id_H .	Main dependency
S3-T2	Scale representatives are useful but not primitive anchors.	Theorem
S3-T3	Every standing-bearing scale claim factors through R_H^{scale} with $\text{NF}_H^{\text{stand}}$ defined.	Normal-form theorem
S3-T4	The raw-role and standing-status classifiers are total on submitted same-scope claims.	Classifier theorem
S3-T5	No primitive Higgs scale survives.	Main endpoint
S3-T6	False scale anchors reduce to raw roles plus standing-status outcomes: carrier-indexed, standing-smuggling, target update, or failed primitive.	Exhaustion theorem
S3-T7	Carrier comparison precedes metric ratio display.	Carrier-order theorem
S3-T8	Metric projection can preserve but not generate carrier order.	Projection theorem
S3-T9	Conditional faithful hierarchy projection displays C_H on the interior/subordinate side when the carrier-order premise is declared.	Conditional orientation theorem
S3-N1	The paper does not derive v_{EW} .	Nonclaim
S3-N2	The paper does not derive m_H .	Nonclaim
S3-N3	The paper does not derive M_{Pl} or $\overline{M}_{\text{Pl}}$.	Nonclaim
S3-N4	The paper does not solve the numerical hierarchy problem.	Nonclaim
S3-N5	The paper does not rule out UV completions or BSM mechanisms.	Nonclaim
S3-N6	The paper does not deny ordinary scale measurements or RG calculations.	Nonclaim
S3-N7	The paper does not independently derive the carrier-order premise $C_H \triangleleft C_G^{\text{env}}$.	Nonclaim

K Source and dependency ledger

Table 13: Source and dependency ledger.

Source family	Imported role	Local use
Kernel and admissibility structure	Fixed-domain kernel, AMetricity, bivalence, standing conservation	Background proof class, no same-scope primitive promotion.
Fixed-domain exhaustion and closure by exhaustion	No fifth governance-bearing locus, same-scope operator closure	Scale classifier totality and no-additional-class result.
Paper 1: Higgs Bridge Carrier	$D_H, C_H, H_{\text{role}}, \mathcal{R}_H, R_H, \mathbf{SF}_H$	Carrier-indexed scale normal form.
Paper 2: Higgs Structural Identity	$\text{Id}_H = \text{Occ}_{D_H}(C_H)$	Prevents scale claims from reopening identity.
Scale Standing and Higgs-Metric Envelope	False-anchor elimination and carrier/projection hierarchy posture	Hierarchy orientation and metric projection sections.
No Primitive Lower Higgs Vacuum	Lower-branch and instability-scale non-promotion; route-status discipline	Handoff to vacuum routes and instability-scale treatment.
Standard electroweak/QFT background	Higgs field, vev, pole mass, RG, matching, effective potential	Provides representative language; not used as primitive standing authority.

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